A Novel Physics-Based Empirical Line Approach for Reflectance Calibration of UAV Multispectral Imagery

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***Abstract*—Accurate reflectance estimation from uncrewed aerial vehicle (UAV) multispectral imagery is critical for a wide range of applications and is achieved using various calibration methods. The conventional empirical line method (ELM) is a statistical digital number (DN) calibration approach that ignores input irradiance, which can introduce reflectance errors, whereas more physically accurate approaches require additional upward-looking irradiance sensors and separate calibration procedures. This study presents a novel reflectance formulation that bridges physical reflectance modeling and statistical DN calibration by reconstructing effective irradiance directly from reference panels, eliminating the need for an irradiance sensor. The method does not require additional irradiance sensors on the UAV or ground and was evaluated under variable illumination conditions. The proposed workflow first extracts raw DNs from UAV multispectral imagery together with four ground reference panels of known reflectance to establish the physical relationship between surface reflectance, radiance, and irradiance. The reference panels are measured at the start or end of the flight, or during UAV overpass acquisition. The certified spectral library of the reference calibration panels provides physically reliable reference reflectance values, which are incorporated into the radiometric model to reconstruct the effective irradiance and preserve the physical irradiance–reflectance relationship throughout the calibration process. Finally, the calibration coefficients are solved using a weighted least-squares optimization to estimate the effective irradiance. The validation results demonstrate that the proposed method significantly improves reflectance accuracy (R^2 = 0.96) and robustness compared with the conventional ELM. The proposed method reduced the root mean square error (RMSE) by approximately 50% compared with the conventional ELM approach. Overall, the proposed method offers a practical calibration approach for UAV-based reflectance estimation without requiring an additional irradiance sensor.**

***Index Terms*—Empirical line method, irradiance estimation, multispectral imagery, physics-based calibration, reflectance calibration, uncrewed aerial vehicle (UAV).**

I. Introduction

UNMANNED aerial vehicle (UAV)-based remote sensing is widely used across Earth science disciplines such as oceanography and geology, as well as in application-oriented fields including mineral exploration and precision agriculture [1], [2]. In various applications, surface reflectance is preferred over raw digital numbers (DNs) to enable reliable comparison of imagery acquired at different times. Variations in solar illumination, cloud cover, and shadow conditions can significantly affect image brightness, making direct comparison of DNs unreliable between different image captures [3], [4]. Reflectance correction mitigates these effects by normalizing images acquired under different illumination conditions, ensuring that observed differences are attributable to actual crop changes rather than lighting variability [5], [6]. Recent studies have shown that radiometric calibration remains one of the major challenges in UAV multispectral imaging, particularly under changing illumination conditions [4], [7], [8]. Previous studies have reported that changing illumination conditions caused by cloud movement, varying solar angles, and long UAV flight durations can reduce the accuracy of empirical calibration methods, particularly those assuming constant illumination throughout image acquisition [8], [9]. In contrast, irradiance-based approaches are generally more robust under variable illumination but require additional sensing hardware, additional calibration procedures, and increased operational complexity [9], [10], [11]. Therefore, improving calibration robustness while preserving the physical consistency of the radiometric model remains an active research topic.

In spectral imaging remote sensing, absolute reflectance retrieval methods are generally classified into three categories. The first category comprises traditional empirical line method (ELM), which rely on reference calibration targets and have been widely adopted in UAV-based applications due to their simplicity and operational efficiency [12], [13],

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[14], [15]. The second category includes atmospheric radiative transfer model (ARTM)-based approaches, such as 6S and MODTRAN, which explicitly model atmospheric scattering and absorption processes and are primarily used in satellite remote sensing [16], [17], [18]. The third category involves measuring incident radiation (MIR) using dedicated irradiance sensors synchronized with image acquisition, which is particularly suitable for UAV and close-range remote sensing [9], [10], [11]. However, these methods require additional calibration procedures and increase system complexity and operational cost. Recent studies have further explored real-time reflectance generation by integrating onboard downwelling irradiance measurements into UAV calibration workflows to better compensate for illumination variability during image acquisition [19]. Although these approaches improve radiometric consistency, they still depend on dedicated irradiance sensors. In contrast, the proposed method reconstructs an effective irradiance directly from reference-panel observations together with spectral library information, eliminating the need for additional irradiance sensing hardware.

Among these approaches, the ELM, a statistical DN calibration approach, is one of the most widely used techniques for reflectance calibration in UAV multispectral imagery [20], [21]. The classical ELM uses ground reference panels with known reflectance values to establish a linear relationship between image DNs and surface reflectance, assuming stable illumination conditions during the flight [20]. In practice, some ELM implementations use a single calibration target, while others employ two or more targets to improve calibration robustness. In all cases, a linear calibration equation is independently derived for each sensor band under the assumption of linearity between DN and reflectance [12], [15], [21], [22]. Recent comparative studies have shown that multipanel ELM generally provides higher calibration accuracy than single-panel approaches, particularly under overcast or variable illumination conditions [8].

The accuracy of the ELM is strongly dependent on coefficient estimation. Several studies have shown that the relationship between DNs and surface reflectance is not always strictly linear, particularly under complex illumination and atmospheric conditions. To address these issues, nonlinear and physics-based calibration models have been proposed to better represent the underlying radiometric processes and improve reflectance retrieval accuracy beyond the conventional ELM framework [21], [23], [24]. These findings highlight the importance of explicitly considering incoming irradiance, which is implicitly assumed to be constant and therefore ignored in conventional ELM formulations.

Spectral libraries contain calibrated reflectance measurements obtained under controlled conditions, providing consistent reference values for radiometric calibration [25], [26]. In this study, the spectral library does not refer to a vegetation or publicly available spectral database; instead, it consists of certified reflectance spectra acquired from the reference calibration panels used throughout the calibration process. Therefore, they provide stable, repeatable, and traceable reference reflectance values that are independent of individual UAV flights and illumination conditions, making them more reliable than image-derived observations for calibration. Furthermore, spectral libraries provide an effective way to bridge physically based radiative formulations with the statistical ELM. This concept complements recent irradiance-based reflectance generation methods [19], while the proposed framework estimates the effective irradiance directly from reference-panel observations and spectral library information without requiring an onboard irradiance spectrometer. Recent studies have also highlighted the importance of performing calibration under actual flight conditions and improving the physical consistency of radiometric calibration workflows [27]. By using known reflectance values from spectral libraries together with multiple ground reference panels, irradiance can be treated as an unknown but estimable variable rather than a fixed assumption. This allows the physical irradiance–reflectance relationship to be preserved within the calibration process, leading to improved robustness against illumination variability and enhanced physical consistency across different flights and environmental conditions.

The objectives of this study are to

1) develop a novel physics-based empirical line approach to robustly estimate calibration coefficients and enhance the physical foundation of reflectance calibration;
2) estimate input irradiance using spectral library data together with multiple ground reference panels; and
3) finally, apply a weighted optimization strategy to stabilize calibration coefficients and improve robustness under variable illumination conditions.

To achieve these objectives, a physics-based empirical line framework is proposed by integrating the certified spectral library of reference calibration panels with effective irradiance reconstruction. The reconstructed irradiance is incorporated into the calibration model to estimate physically consistent calibration coefficients. The proposed method is evaluated under different illumination conditions and validated using field spectroradiometer measurements and vegetation indices. Its performance is finally compared with the conventional ELM. The proposed framework reconstructs a single scene-wide effective irradiance for each calibration acquisition and employs it uniformly for the radiometric calibration of the corresponding mosaic.

II. Materials and Methods

The overall workflow of the proposed method is illustrated in Fig. 1. First, multispectral UAV images are acquired together with four ground reference calibration panels. Four reference panels with different reflectance levels were selected to provide a broad reflectance range for robust calibration coefficient estimation. The certified spectral library provides accurate reference reflectance values for the calibration panels, while the corresponding image DNs are extracted from the UAV imagery. These data are integrated into the proposed physics-based model to reconstruct the effective irradiance and estimate robust calibration coefficients using a weighted least squares (WLSs) optimization. The estimated coefficients are then used to convert image DNs into surface reflectance. The proposed method is subsequently evaluated under different illumination conditions, flight altitudes, viewing angles,

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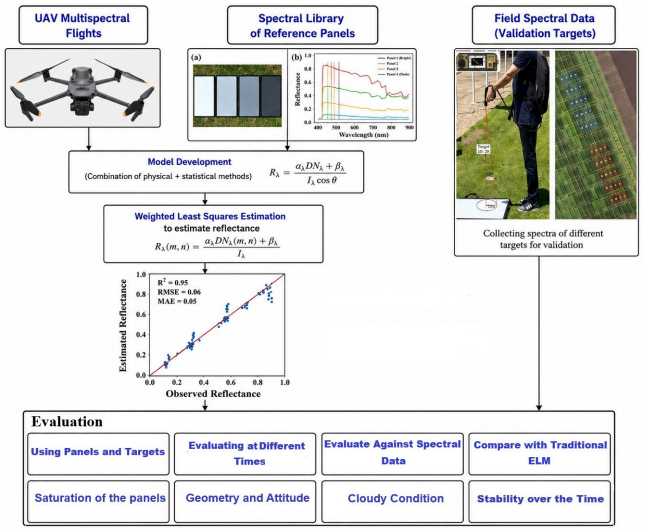


Fig. 1. Workflow of the physics-based empirical line approach for reflectance calibration. (a) Four reference calibration panels and (b) spectral reflectance curves of the four panels.

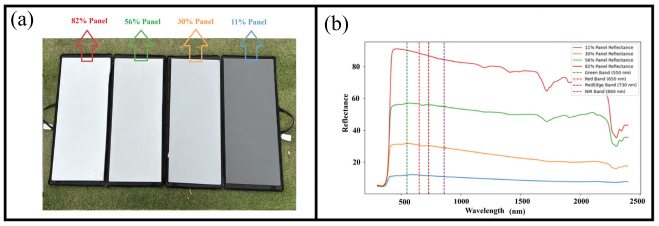


Fig. 2. (a) Four GRYFN aluminum di-bond reference panels with nominal reflectance values of 11%, 30%, 56%, and 82%. (b) Spectral reflectance of the panels, showing their stability across the visible to NIR range (420–2500 nm) and the spectral positions of the M3M bands (green: 560 nm, red: 650 nm, RE: 730 nm, NIR: 860 nm).

acquisition times, and panel saturation scenarios. It is then validated using field spectroradiometer measurements collected over natural targets, and its practical performance is further assessed using vegetation indices. Two phenological stages were selected to evaluate the calibration method under different canopy conditions. Spectra of ground targets were collected from approximately 120 cm above the surface to obtain pure spectra based on the sensor field of view (FOV), whereas canopy spectra were collected from approximately 30 cm above the canopy following the Australian Plant Phenomics Network (APPN) protocol. The weather conditions during image acquisition have been added to the revised manuscript. Depending on the experiment, the measurements were conducted under clear, cloudy, or overcast conditions to evaluate the robustness of the proposed method, with light wind (less than 5 km·h^{-1}). No rain or tree shadows were present during the measurements, and the reference panels were placed in an open area to ensure uniform illumination throughout the experiments. Finally, the results are compared with those obtained using the conventional ELM to demonstrate the accuracy, robustness, and practical advantages of the proposed method.

A. Camera and Panels

In this study, a DJI Mavic 3 Multispectral (M3M) UAV system was used for image acquisition. The system integrates a 20 MP RGB camera (4/3 CMOS sensor with a mechanical shutter) and four 5 MP multispectral sensors that simultaneously capture imagery in the Green, Red, Red Edge (RE), and near-infrared (NIR) spectral regions. The multispectral module consists of four discrete spectral bands with the following central wavelengths and bandwidths: green (G), 560 ± 16 nm; red (R), 650 ± 16 nm; RE, 730 ± 16 nm; and NIR, 860 ± 26 nm. The RGB camera provides high-resolution true-color imagery using a mechanical shutter and supports 0.7 s interval shooting, enabling continuous image acquisition during flight. The platform can map up to 200 ha per flight with a maximum flight time of approximately 43 min, making it well suited for agricultural and environmental monitoring applications [28].

Four rigid aluminum di-bond reference panels (GRYFN, USA) were used for radiometric calibration. GRYFN offers a custom spectral reflectance panel deployment kit designed for easy in-field deployment of spectral reflectance panels, which are commonly utilized in hyperspectral and multispectral data collection as a calibration method for converting radiance data captured by a sensor to analysis-ready reflectance data. The GRYFN standard Spectral Reflectance Panel Kit consists of two carrying cases (1.2 × 0.6 × 0.035 m), each containing two rigid di-bond panels (1.10 × 0.54 m) coated with an actinically stable and waterproof fluoropolymer paint. Each panel is painted with a different reflectance percentage of 11%, 30%, 56%, and 82% (high-reflectance white panel), achieved through a pigmented Type UF200 fluoro-urethane polymer coating that maintains spectral stability under prolonged environmental exposure.

The coating provides a matte reflecting surface with a band-averaged diffuse hemispherical reflectivity (DHR) across the visible, NIR, and shortwave-infrared domains (420–2500 nm). Spectral DHR data were measured using a Perkin-Elmer Lambda 1050 spectrophotometer with a 150-mm integrating sphere relative to a Spectralon (Labsphere, Inc.) reference and extended from 2000 to 3000 nm using a Nicolet iS50 FTIR spectrophotometer with a Pike integrating sphere. The certified spectral reflectance values were provided by the manufacturer as part of the panel certification documentation. These reference panels were used to calibrate UAV imagery by converting sensor radiance into analysis-ready surface reflectance. Fig. 2(a) shows the panels, and Fig. 2(b) illustrates the spectral reflectance of the four panels corresponding to the green, red, RE, and NIR spectral bands.

B. Proposed Physics-Based Empirical Line Approach

The proposed calibration model consists of the following steps.

1) Basic Reflectance Model: Surface reflectance (R_λ) is derived from the physical relationship between spectral radiance (P_λ, W·m^{-2} · sr^{-1}·nm^{-1}) and incident spectral irradiance (I_λ, W·m^{-2}·nm^{-1}) for each spectral band (λ), following the classical radiance–reflectance relationship widely used in

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remote sensing [29], [30]. In this study, P_λ denotes the spectral radiance measured at the multispectral sensor, whereas I_λ represents the incident spectral irradiance measured during image acquisition

$$R_\lambda = \frac{\pi P_\lambda d^2}{I_\lambda \cos\theta} \tag{1}$$

where d is the normalized Earth–Sun distance (dimensionless), θ is the solar zenith angle, and R_λ is the dimensionless surface reflectance.

For low-altitude UAV image acquisition, atmospheric effects are assumed to be negligible because of the short sensor-to-target path length and the relatively stable atmospheric conditions during a single flight. Therefore, the atmospheric transmittance term is omitted from the proposed formulation. Furthermore, if the remaining constant terms, π and the Earth–Sun distance (d), are grouped into a single constant (C), this constant remains unchanged throughout a single UAV flight and is therefore absorbed into the proportionality constant without affecting the physical interpretation of the model. Consequently, (1) can be simplified to

$$R_\lambda \propto \frac{P_\lambda}{I_\lambda \cos\theta}. \tag{2}$$

2) Modeling Radiance From DN: Sensor radiance is linearly related to the recorded DN

$$P_\lambda = \alpha_\lambda DN_\lambda + \beta_\lambda. \tag{3}$$

Substituting into the reflectance equation gives

$$R_\lambda = \frac{\alpha_\lambda DN_\lambda + \beta_\lambda}{I_\lambda \cos\theta}. \tag{4}$$

To compute irradiance accurately and simplify the formulation, an effective irradiance is defined as

$$I_\lambda^c = \frac{I_\lambda \cos\theta}{\pi}. \tag{5}$$

Thus, the practical equation becomes

$$R_\lambda = \frac{\alpha_\lambda DN_\lambda + \beta_\lambda}{I_\lambda^c}. \tag{6}$$

This forms the core empirical relationship between the image DN and surface reflectance.

3) Reference-Panel Calibration System: Four ground reference panels with known reflectance values R_λ^i are used for calibration. For each panel $i = 1, \ldots, 4$

$$R_\lambda^i I_\lambda^c = \alpha_\lambda DN_\lambda^i + \beta_\lambda. \tag{7}$$

This yields an over-determined system with three unknowns $(\alpha_\lambda, \beta_\lambda, I_\lambda^c)$.

4) Initial Estimation (Assuming $I_\lambda^c = 1$): Initially, $I_\lambda^c = 1$, this initial value is assumed to be unity solely for initialization of the iterative optimization. It is subsequently estimated from the fit panel observations and updated during the calibration process. Therefore, the simplified system becomes

$$R_\lambda^i = \alpha'_\lambda DN_\lambda^i + \beta'_\lambda. \tag{8}$$

The least squares solution is

$$X = \left(A^T A\right)^{-1} A^T L \tag{9}$$

$$\mathbf{A} = \begin{bmatrix} DN_\lambda^1 & 1 \\ DN_\lambda^2 & 1 \\ DN_\lambda^3 & 1 \\ DN_\lambda^4 & 1 \end{bmatrix} \tag{10}$$

$$\mathbf{L} = \begin{bmatrix} R_\lambda^1 \\ R_\lambda^2 \\ R_\lambda^3 \\ R_\lambda^4 \end{bmatrix} \tag{11}$$

$$\mathbf{X} = \begin{bmatrix} \alpha'_\lambda \\ \beta'_\lambda \end{bmatrix}. \tag{12}$$

5) Reconstruction of Effective Irradiance and Final Coefficients: Rearranging (7) gives an individual estimate of the effective irradiance for each reference panel

$$I_{\lambda,i}^c = \frac{\alpha'_\lambda DN_\lambda^i + \beta'_\lambda}{R_\lambda^i}. \tag{13}$$

Since all reference panels are imaged under the same illumination conditions, the effective irradiance should theoretically be identical for all panels. Therefore, the final effective irradiance is estimated as the arithmetic mean of the individual panel estimates

$$I_\lambda^c = \frac{1}{n} \sum_{i=1}^{n} I_{\lambda,i}^c. \tag{14}$$

This averaging provides a stable scene-level estimate of the effective irradiance by reducing the influence of sensor noise, panel-specific measurement uncertainty, and local illumination variability.

Based on the physical–empirical formulation in (6), the reflectance for each calibration panel can be expressed as

$$R_\lambda^i = \frac{\alpha_\lambda DN_\lambda^i + \beta_\lambda}{I_\lambda^c}. \tag{15}$$

Rearranging (15) gives

$$R_\lambda^i = \left(\frac{\alpha_\lambda}{I_\lambda^c}\right) DN_\lambda^i + \left(\frac{\beta_\lambda}{I_\lambda^c}\right). \tag{16}$$

This expression has the same form as the empirical linear relationship given in (8)

$$R_\lambda^i = \alpha'_\lambda DN_\lambda^i + \beta'_\lambda. \tag{17}$$

Comparing (16) and (17) yields

$$\alpha'_\lambda = \frac{\alpha_\lambda}{I_\lambda^c}, \qquad \beta'_\lambda = \frac{\beta_\lambda}{I_\lambda^c}. \tag{18}$$

The coefficients α'_λ and β'_λ obtained from the initial WLS solution are normalized coefficients estimated with the initialization $I_\lambda^c = 1$. They are intermediate coefficients rather than the final physical calibration coefficients. Accordingly, the final calibration coefficients are recovered as

$$\alpha_\lambda = \alpha'_\lambda I_\lambda^c, \qquad \beta_\lambda = \beta'_\lambda I_\lambda^c. \tag{19}$$

To improve robustness, a WLSs method is applied with weight w_i assigned to each reference panel. The final weighted solution is

$$X = \left(A^T W A\right)^{-1} A^T W L \tag{20}$$

$$W = \mathrm{diag}\left(w_1, w_2, \ldots, w_n\right). \tag{21}$$

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The weights are computed as

$$w_i = \frac{1}{e_i^2 + \epsilon} \tag{22}$$

where ϵ is a small regularization constant introduced to avoid division by zero. The residual for each panel is defined as

$$e_i = R_\lambda^i - \left(\alpha'_\lambda DN_\lambda^i + \beta'_\lambda\right). \tag{23}$$

6) Pixel-Wise Reflectance Calculation: Using the optimized parameters, reflectance for each image pixel (m, n) is computed as

$$R_\lambda(m,n) = \frac{\alpha_\lambda DN_\lambda(m,n) + \beta_\lambda}{I_\lambda^c}. \tag{24}$$

This equation is applied to all bands, producing a radiometrically corrected reflectance mosaic that is more consistent across different illumination conditions and acquisition times.

C. Comparison of Computed Reflectance With Observed Reflectance Derived From Spectrometer

To validate the accuracy of the proposed reflectance calibration method, field spectra were collected during UAV image acquisition using a Spectra Vista Corporation (SVC) HR-1024i spectroradiometer (SVC, USA). The instrument operates over a spectral range of 350–2500 nm with an instantaneous FOV of 25° and provides high-resolution radiometric measurements, with spectral sampling of approximately 1.5 nm in the VNIR and 3.8 nm in the SWIR regions.

The recorded SVC spectra were resampled to match the spectral response functions (SRFs) of the UAV multispectral camera. For each multispectral band, the SVC reflectance was computed as the mean reflectance within the corresponding band spectral range, weighted by the camera's SRF. This ensured spectral consistency and wavelength compatibility between the ground-based and UAV-derived reflectance measurements.

In order to evaluate the performance of the proposed radiometric calibration method under realistic field conditions, reflectance values estimated from the M3M imagery were first compared with SVC field spectroradiometer measurements acquired at 15-min intervals. In addition, to provide an independent validation, one gray M3M panel (different from the four Griffin calibration panels) and three natural targets (asphalt, sand, and grass) were measured using both the SVC field spectroradiometer and the M3M multispectral camera at the same intervals, as illustrated in Fig. 3. Field spectral measurements were acquired at a sensor height of 120 cm above the targets, corresponding to a ground footprint radius of approximately 26 cm, as shown in Fig. 3.

Measurements were collected at four different times at 15-min intervals to capture natural variations in illumination and assess the temporal stability of the proposed method. It is worth noting that target reflectance was measured during each flight. At each time point, the same eight targets were measured within a short acquisition interval using both the SVC spectrometer and the M3M multispectral camera, resulting in a total of 32 paired measurements (eight targets × four acquisition times).

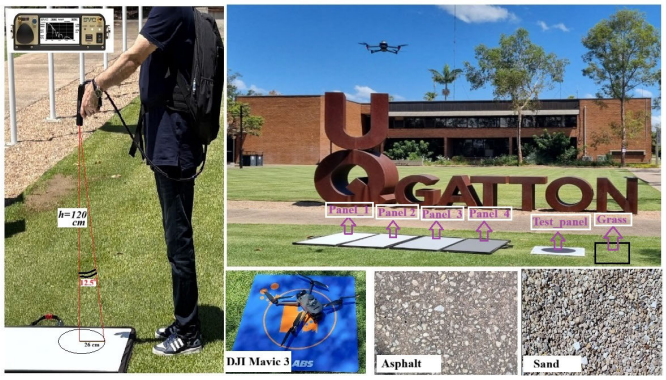


Fig. 3. Spectral collection of calibration panels using the SVC spectroradiometer with a 25° FOV, and comparison with DJI M3M images over reference panels, gray panel, asphalt, sand, and grass. To provide an independent validation of the proposed method, the estimated reflectance values in different spectral bands were compared with SVC measurements acquired over independent surfaces, including gray panel, sand, asphalt, and grass, which were not used for calibration. The SVC measurements were collected within a short time window (approximately 3 min) immediately after the UAV flight to ensure consistent illumination conditions. No noticeable shadow effects were observed in the UAV imagery, as the ground measurements were conducted outside the UAV acquisition period.

For each acquisition, SVC spectra and M3M images were collected under similar illumination conditions to ensure a fair comparison. The mean reflectance derived from the SVC measurements was then compared with the reflectance estimated by the M3M calibration model. The agreement between the two datasets was quantified using the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE), providing a comprehensive evaluation of the accuracy, consistency, and robustness of the proposed calibration method across different surface types and varying illumination conditions.

D. Derived Vegetation Index Values Using M3M and Field Spectrometry

To evaluate the performance of the proposed physics-based calibration method under field conditions typical of agricultural systems, three vegetation indices, including the normalized difference vegetation index (NDVI), simple ratio (SR), and green NDVI (GNDVI), were selected for validation, as shown in Table I. These indices were selected because they represent different spectral responses based on the available green, red, and NIR bands, allowing the proposed calibration method to be evaluated under different types of spectral variability. Selected indices are sensitive to radiometric errors, making them suitable for evaluating the accuracy and consistency of the reflectance calibration. Indices were calculated from the calibrated reflectance values produced by the proposed method and compared with the corresponding reflectance values measured directly by the field spectrometer to assess accuracy and consistency across spectral bands.

Since the main application of the M3M sensor is in the agricultural domain, its performance was evaluated using a controlled field experiment conducted in Australia at a research field located in Gatton (Queensland). As shown in Fig. 4, a total of 42 plots were included in this study, consisting

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TABLE I
SPECTRAL VEGETATION INDICES USED IN THIS STUDY

Vegetation Index	Formula	Explanation	Reference
SR	$\frac{\text{NIR}}{\text{Red}}$	Simple Ratio	[31]
NDVI	$\frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$	Normalized Difference Vegetation Index	[32]
GNDVI	$\frac{\text{NIR}-\text{Green}}{\text{NIR}+\text{Green}}$	Green Normalized Difference Vegetation Index	[33]

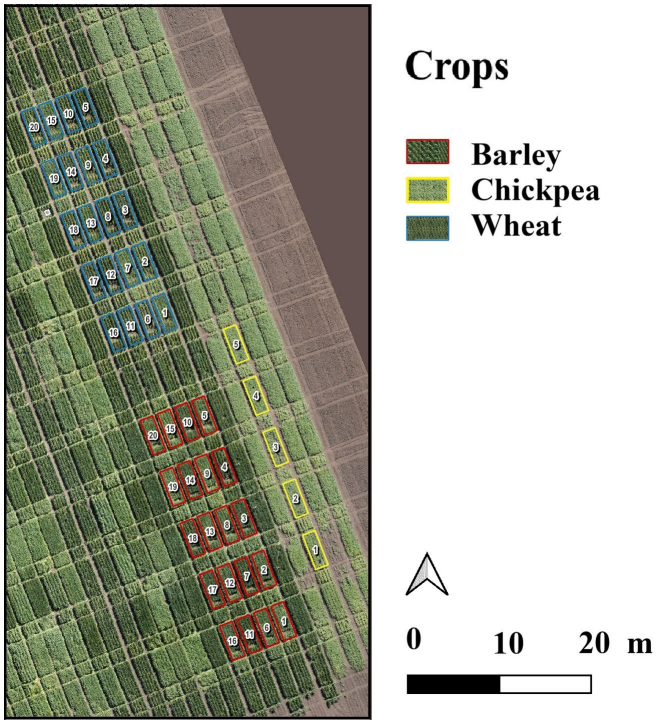


Fig. 4. Location of the plots used for measuring wheat, barley, and chickpea reflectance using the SVC spectrometer and the M3M multispectral camera.

of 18 wheat plots, 18 barley plots, and six chickpea plots, with each plot representing a different variety or replicate. Each plot exhibited a distinct canopy structure and leaf area, resulting in species- and variety-specific spectral responses. Spectral measurements were collected at approximately 30 cm above the canopy.

Spectral measurements were collected using the SVC spectroradiometer at two key phenological stages: peak of greenness (PG) and grain filling (GF). At each stage, six spectra were recorded from every plot and averaged to produce one representative reflectance curve, resulting in 84 mean plot-level spectra across the two stages of the growing season. The reflectance values derived from the M3M were then compared with the corresponding SVC-derived spectra for each plot at both measurement times using statistical metrics.

To enable direct comparison with the M3M multispectral camera, both SVC measurements and M3M flights were carried out within a short acquisition interval, ensuring strict temporal alignment. The continuous SVC spectra were preprocessed and resampled to match the M3M spectral bands using the sensor's SRFs.

E. Comparison With the Traditional Empirical Line Method

In this study, the reflectance values obtained using the proposed calibration method were compared with those derived from the traditional four-panel ELM. The conventional ELM establishes a linear relationship between image DNs and the known reflectance values of multiple reference panels for each spectral band, assuming constant illumination conditions during image acquisition.

For the traditional ELM implementation, surface reflectance (R) is computed by fitting a linear regression between the DN values extracted from calibration panels and their corresponding laboratory-measured reflectance values. This relationship is expressed as a linear DN–reflectance model for each spectral band. Once derived, the regression parameters are applied to convert DN values across the image to surface reflectance.

For comparison purposes, the traditional ELM was applied to the same UAV multispectral imagery and calibration panel measurements used in the proposed method. This ensured identical acquisition geometry, illumination conditions, and reference panel properties. A total of 96 observations, collected between 09:30 and 12:00 at six time intervals (30-min spacing) across four spectral bands (green, red, RE, and NIR) and four reference panels (11%, 30%, 56%, and 82%), were used for evaluation.

This experimental design enabled a direct and consistent comparison between the proposed physics-based calibration approach and the conventional empirical line method, isolating methodological differences while controlling for environmental and instrumental factors.

F. Validation Method (Statistical Evaluation)

In this study, the coefficient of determination (R^2), the RMSE, and the MAE were used to evaluate the performance of the proposed calibration method in estimating surface reflectance. The coefficient R^2 quantifies the strength of the linear relationship between calibrated and reference reflectance, while RMSE and MAE measure the magnitude of deviation between them

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_i^{\text{cal}} - R_i^{\text{ref}}\right)^2} \tag{25}$$

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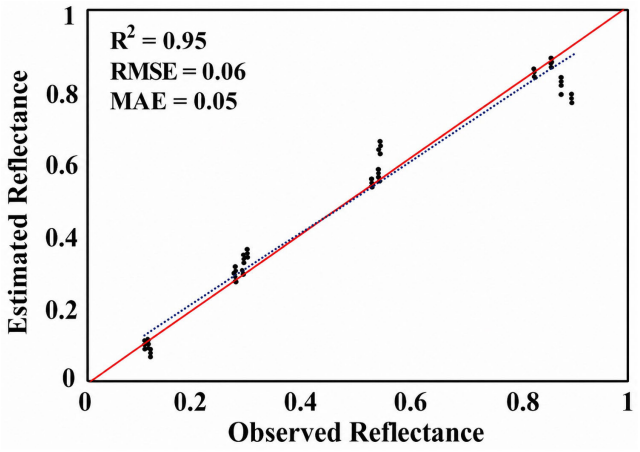


Fig. 5. Accuracy of the proposed method in the estimation of surface reflectance.

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|R_i^{\mathrm{cal}} - R_i^{\mathrm{ref}}\right| \tag{26}$$

$$R^2 = \frac{\left[\sum_{i=1}^{N}\left(R_i^{\mathrm{ref}} - \overline{R^{\mathrm{ref}}}\right)\left(R_i^{\mathrm{cal}} - \overline{R^{\mathrm{cal}}}\right)\right]^2}{\sum_{i=1}^{N}\left(R_i^{\mathrm{ref}} - \overline{R^{\mathrm{ref}}}\right)^2 \sum_{i=1}^{N}\left(R_i^{\mathrm{cal}} - \overline{R^{\mathrm{cal}}}\right)^2}. \tag{27}$$

Here, R_i^{cal} and R_i^{ref} are the calibrated and reference (spectral library) reflectance values for sample i, respectively, and $\overline{R^{\mathrm{cal}}}$ and $\overline{R^{\mathrm{ref}}}$ are their corresponding mean values. Higher R^2 and lower RMSE and MAE indicate better calibration performance and greater consistency with the spectral library reference.

III. Results

A. Validation of the Proposed Model

To evaluate the proposed calibration method, the manufacturer's certified spectral reflectance values of the GRYFN aluminum dibond reference panels (see Fig. 2) were used as the observed reference data. The comparison between the calibrated UAV-derived reflectance and the reference panel reflectance showed a strong linear relationship (R^2 = 0.95, RMSE = 0.06, MAE = 0.05), as shown in Fig. 5. The small difference between RMSE and MAE indicates that the calibration errors are minimal and uniformly distributed across all reflectance levels. A total of 96 observations, collected between 09:30 and 12:00 at six time intervals (at 30-min intervals) across four spectral bands (green, red, RE, and NIR) and four reference panels (11%, 30%, 56%, and 82%), were used for validation. The near 1:1 alignment of the regression line confirms that the proposed calibration approach effectively compensates for illumination variability and sensor-related errors. These results demonstrate that the physics-based ELM combined with weighted least-squares optimization provides a robust and reliable radiometric correction under varying illumination conditions.

The relationship between the mean DNs and the corresponding panel reflectance values for each spectral band (Green, Red, RE, and NIR) showed consistently strong correlations, as illustrated in Fig. 6. The results indicated R^2 = 0.87 for the Green band, R^2 = 0.93 for the Red band, and R^2 = 1.00 for both the RE and NIR bands. The RMSE values ranged from 0.00 (RE and NIR) to 0.10 (Green), while MAE values ranged from 0.00 to 0.09.

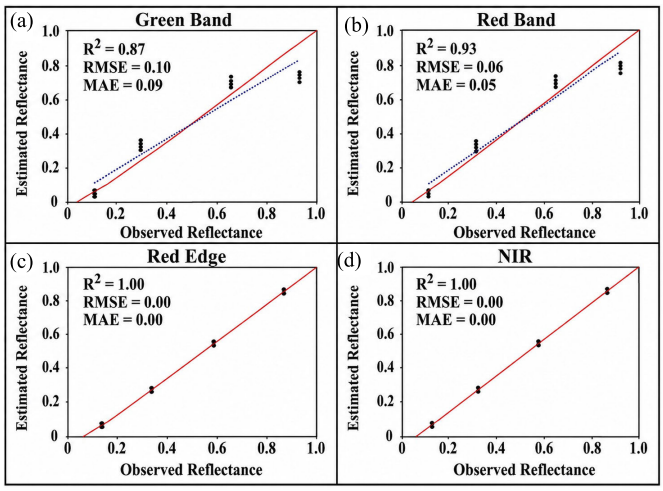


Fig. 6. Accuracy of the proposed calibration method in the estimation of reflectance for each spectral band (Green, Red, RE, and NIR). Estimated versus observed reflectance for (a) green, (b) red, (c) red edge, and (d) NIR bands.

The slightly lower R^2 values observed in the visible bands (Green and Red) are attributed to their higher sensitivity to illumination variability at shorter wavelengths. In contrast, the RE and NIR bands exhibited near-perfect linearity owing to their stronger and more stable reflectance response. The close agreement between RMSE and MAE further indicates that residual errors are minimal and evenly distributed, confirming that the proposed calibration method preserves radiometric integrity and spectral consistency across all multispectral bands.

To evaluate the temporal stability of the proposed illumination correction method, the relationship between estimated and observed reflectance was examined at six different times from 09:30 A.M. to 12:00 P.M. at 30-min intervals, as shown in Fig. 7(a)–(f). The results demonstrate consistently high coefficients of determination (R^2 = 0.94–0.95) with RMSE (0.06) and MAE (0.05), confirming that the method provides stable and accurate reflectance estimates throughout the morning period.

These findings are further supported by the combined results shown in Fig. 7, where all measurements exhibit a strong linear relationship between estimated and observed reflectance. From a physical perspective, this temporal consistency indicates that the correction algorithm effectively compensates for variations in solar illumination, sensor-view geometry, and atmospheric brightness. The minimal variation across time intervals confirms the robustness and reliability of the proposed method for multispectral analyses conducted under changing illumination conditions.

The validation analysis further demonstrated that the proposed physics-based calibration method produces highly accurate reflectance estimates compared with SVC field spectrometer measurements, using independent surfaces (sand,

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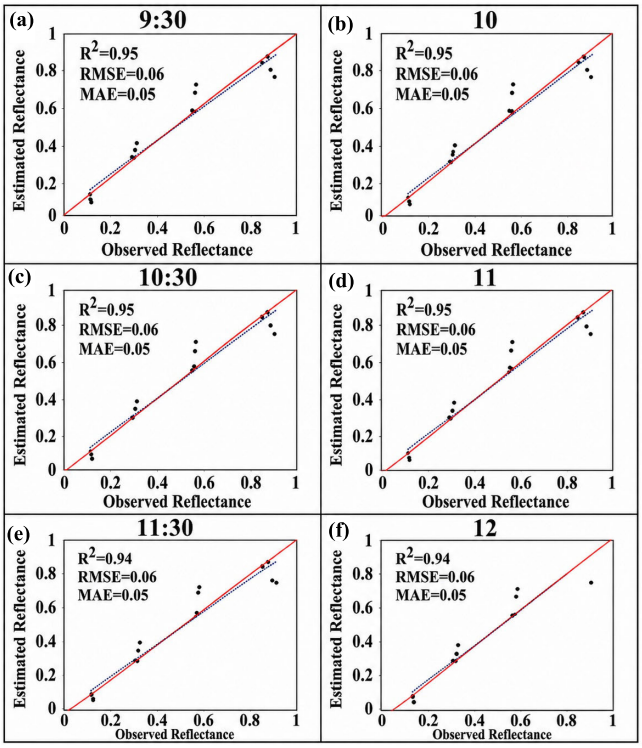


Fig. 7. Accuracy of the proposed calibration method at different times between 09:30 A.M. and 12:00 P.M. at 30-min intervals. Estimated versus observed reflectance at (a) 09:30, (b) 10:00, (c) 10:30, (d) 11:00, (e) 11:30, and (f) 12:00.

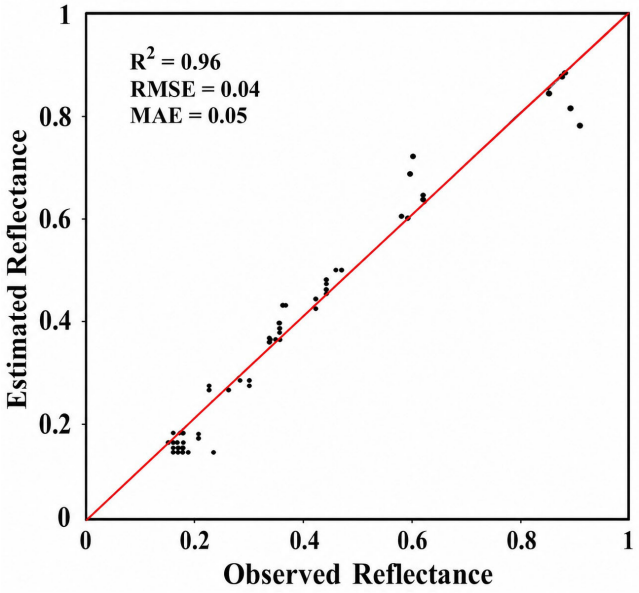


Fig. 8. Comparison of estimated and SVC-measured reflectance for independent targets.

asphalt, grass, and gray panel). As shown in Fig. 8, a strong linear relationship was observed between the M3M-derived reflectance and the SVC-measured reflectance across the evaluated spectral bands. The calibration model achieved an R^2 value of 0.96, indicating excellent agreement between the two datasets. The RMSE (0.04) and MAE (0.05) confirm that the estimated reflectance values closely match the ground-measured spectra, with only minor residual errors. These results verify the accuracy, stability, and practical applicability of the calibration approach developed.

In addition, for completeness, the estimated intermediate calibration parameters are reported for each spectral band. For example, the effective irradiance (I_λ^c) was estimated as 0.448, 0.461, 0.469, and 0.476 for the NIR, RE, Red, and Green bands, respectively. The corresponding normalized coefficients (α'_λ, β'_λ) were (0.590, −0.062), (0.455, −0.089), (0.460, −0.086), and (0.465, −0.213), while the final calibration coefficients (α_λ, β_λ) were (0.264, −0.028), (0.210, −0.041), (0.216, −0.040), and (0.221, −0.101), respectively. These intermediate parameters provide additional insight into the calibration process and may facilitate a better understanding and reproducibility of the proposed method.

B. Changes in Derived Vegetation Indices Using the Proposed Calibration Method Over Time

Using the resampled SVC spectra and the corresponding reflectance derived from the proposed calibration method, three vegetation indices—NDVI, SR, and GNDVI—were calculated for measurement times corresponding to the PG and the GF stages. Fig. 9(a)–(c) illustrates the relationships between the vegetation indices derived from SVC measurements and those calculated directly from the M3M multispectral imagery.

As shown in Fig. 9(a), NDVI achieved an R^2 value of 0.94 with RMSE = 0.05 and MAE = 0.05, indicating excellent consistency between the two datasets. The SR [see Fig. 9(b)] also showed high accuracy, with an R^2 value of 0.93, RMSE = 0.05, and MAE = 0.05. Similarly, GNDVI [see Fig. 9(c)] exhibited a strong relationship, with an R^2 value of 0.83, RMSE = 0.12, and MAE = 0.11.

Together, these results demonstrate that the proposed calibration method performs reliably during both the period of maximum canopy greenness and the GF stage. Overall, the vegetation indices derived from the proposed method closely matched those measured by the field spectrometer at both PG and GF stages. The consistently high coefficients of determination and low error metrics across all indices confirm that the proposed calibration method accurately preserves the spectral relationships required for vegetation index computation and remains robust under changing illumination conditions and phenological stages.

C. Result of Using the Traditional ELM Method

The correlation between reflectance values obtained from the M3M using the traditional ELM and the SVC spectrometer measurements was evaluated across the same observation period. As shown in Fig. 10, the relationship between the ELM-derived reflectance and the SVC-measured reflectance across the evaluated spectral bands yielded an R^2 value of 0.88. This indicates a weaker correlation compared with the proposed physics-based calibration method, which achieved an R^2 value of 0.96 (see Fig. 8). In addition, the traditional

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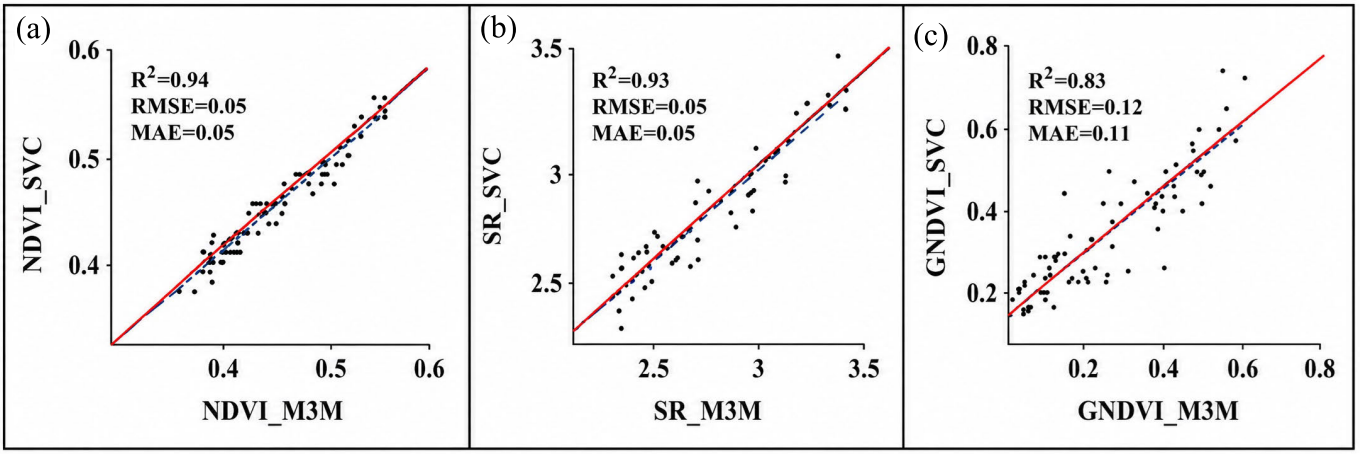


Fig. 9. Accuracy of the proposed calibration method in the estimation of (a) NDVI, (b) SR, and (c) GNDVI using SVC spectrometer data and the M3M multispectral camera for wheat, barley, and chickpea.

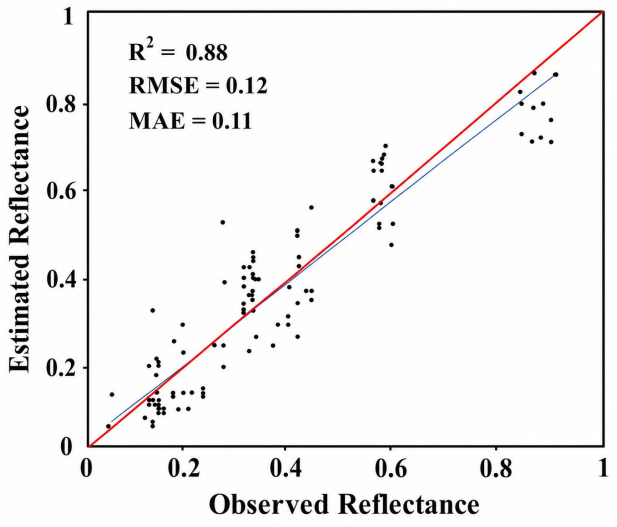


Fig. 10. Accuracy of the traditional ELM method in the estimation of surface reflectance using SVC spectrometer data.

ELM produced RMSE = 0.12 and MAE = 0.11. In comparison, the proposed physics-based calibration method yielded substantially lower errors, with RMSE = 0.04 and MAE = 0.05. The fit regression deviates below the 1:1 line, indicating a slight underestimation of reflectance by the traditional ELM-based single-panel calibration method. This behavior contrasts with the proposed physics-based calibration approach (see Fig. 8), in which the fit regression closely overlaps the 1:1 line. These results highlight the superior performance and robustness of the proposed calibration method for accurate reflectance estimation under variable illumination conditions.

D. Evaluation Under Different Elevation and Viewing Angle Conditions

To evaluate the effect of altitude on the model performance, the proposed method was tested at four altitudes (5, 10, 20, and 40 m). Each altitude was repeated five times to obtain a reliable estimation.

TABLE II
MODEL PERFORMANCE AT DIFFERENT ALTITUDES

Altitude (m)	R^2	RMSE	MAE
5	0.96	0.05	0.05
10	0.96	0.05	0.05
20	0.94	0.06	0.06
40	0.94	0.06	0.06

TABLE III
MODEL PERFORMANCE AT DIFFERENT VIEWING ANGLES

Viewing Angle (°)	R^2	RMSE	MAE
0	0.96	0.05	0.05
10	0.96	0.05	0.05
20	0.96	0.05	0.05
30	0.94	0.06	0.06

As summarized in Table II, the model shows consistent performance across all elevations. The R^2 values range from 0.96 to 0.94, while RMSE and MAE vary slightly between 0.05 and 0.06. The highest performance is observed at lower altitudes (5 and 10 m), with $R^2 \approx 0.96$. At 20 m, the performance remains strong with only a slight decrease, while at 40 m, a minor reduction is observed.

These results indicate that the model is stable across different altitudes, with only a small decline at higher altitudes, likely due to atmospheric effects, BRDF, and pixel mixing. The use of reference panels helps to reduce illumination and atmospheric variability.

To further assess the effect of viewing geometry, the model was evaluated at four different angles (0°, 10°, 20°, and 30°) while maintaining a fixed altitude of 40 m. Each experiment was repeated three times to ensure consistency.

As shown in Table III, the model performance remains nearly identical at 0°, 10°, and 20°, with $R^2 \approx 0.96$ and RMSE and MAE around 0.05. At 30°, a slight decrease in performance is observed, where R^2 reduces to approximately 0.95.

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TABLE IV
TEMPORAL STABILITY OF THE PROPOSED METHOD DURING UAV FLIGHT

Time Interval (minutes)	R^2	RMSE	MAE
10	0.96	0.05	0.05
15	0.94	0.06	0.05
20	0.93	0.06	0.06
30	0.92	0.07	0.06

These findings indicate that the model is largely insensitive to viewing angle, with only a minor reduction in accuracy at higher angles, confirming the robustness of the proposed method against angular effects.

E. Stability of the Proposed Method Over Time

To evaluate the temporal stability of the proposed method, three test runs were conducted using the M3M system, considering a maximum flight duration of approximately 30 min. The experiments were performed at different times: from 10:00 to 10:30, 11:00 to 11:30, and 12:00 to 12:30.

The results were analyzed to assess the effect of time on model performance. As shown in Table IV, the model accuracy remained stable at approximately $R^2 \approx 0.96$ for time intervals up to 10 min. However, the accuracy gradually decreased to about $R^2 \approx 0.94$ at 15 min and further to 0.93 and 0.92 at 20 and 30 min, respectively.

This behavior is attributed to temporal changes in illumination conditions, which affect the energy captured by the reference panels and, consequently, the radiance estimation derived from DN values. As a result, the accuracy of the model decreases over time.

These findings indicate that recalibration using reference panels at shorter time intervals (approximately 10–15 min) is necessary to maintain high accuracy. The proposed method estimates a single scene-wide effective irradiance for each calibration acquisition and applies it uniformly across the corresponding mosaic. Therefore, it is intended for discrete calibration snapshots rather than continuous frame-by-frame illumination correction during flight. Based on the temporal stability results presented in Table IV, recalibration is recommended approximately every 10–15 min under changing illumination conditions. Unlike interpolation-based in-flight approaches that continuously update illumination during flight, the proposed framework reconstructs the effective irradiance only once for each calibration acquisition and subsequently applies it uniformly to the entire mosaic.

Overall, this strategy provides a practical balance between operational efficiency and model accuracy.

IV. DISCUSSION

A. Evaluation of the Stability and Accuracy of the Proposed Calibration Method

The results demonstrated that the proposed physics-based calibration method provides high stability and accuracy across different illumination conditions and spectral bands. The strong correspondence between SVC-measured and UAV-derived reflectance (R^2 = 0.94–0.95; RMSE = 0.05–0.06; MAE = 0.05–0.06) confirms that the method can reliably correct radiometric distortions without introducing systematic bias. The near 1:1 alignment observed in all regression plots reflects the effectiveness of the weighted least-squares optimization in compensating for panel reflectance variability, sensor noise, and angular effects. Temporal evaluations from 09:30 to 12:00 further showed that the corrected reflectance remained consistent over time, demonstrating that the method is robust and stable even under changing solar elevation. Compared with recent UAV reflectance generation approaches that rely on onboard downwelling spectrometers [19], the proposed framework estimates effective irradiance from reference-panel observations and spectral library information, providing a panel-physics-based extension of the traditional ELM. These findings are consistent with previous studies reporting that variable illumination can affect the accuracy of empirical calibration methods that assume constant illumination during image acquisition. The proposed framework addresses this limitation by estimating effective irradiance from certified panel reflectance values and multiple reference panels.

It should be noted that the reference reflectance values used in Figs. 5–7 were obtained from the manufacturer's certified spectral reflectance library of the GRYFN aluminum dibond reference panels. Consequently, the small shifts observed along the x-axis among the spectral bands are expected because the certified reflectance values vary slightly with wavelength, as illustrated in Fig. 2(b). These wavelength-dependent differences are inherent to the reference panels and do not represent calibration errors.

The improved accuracy observed in this study results from the proposed physics-based calibration framework, which combines the physical radiance–reflectance relationship with effective irradiance estimation and WLSs optimization. By incorporating physically meaningful constraints into the calibration process, the proposed method improves radiometric accuracy and provides greater robustness under varying illumination conditions.

B. Implications for Vegetation Index Retrieval and Crop Monitoring

Accurate estimation of vegetation indices across different growth stages and crop types is essential for quantifying crop conditions, as vegetation indices respond differently to phenological development, canopy structure, and spectral band characteristics [34], [35]. Across three crops and two phenological stages, the vegetation indices derived from both SVC measurements as a reliable reference for validation, and the proposed calibration method exhibited consistent and stable behavior, demonstrating the robustness of the developed approach. Strong agreement was observed between vegetation indices calculated from the M3M imagery and those derived from resampled SVC spectra at both the PG and GF stages. This consistency indicates that the proposed calibration method preserves the essential spectral relationships required for reliable vegetation index computation.

NDVI showed excellent performance across all crops and both growth stages, with high coefficients of determination and

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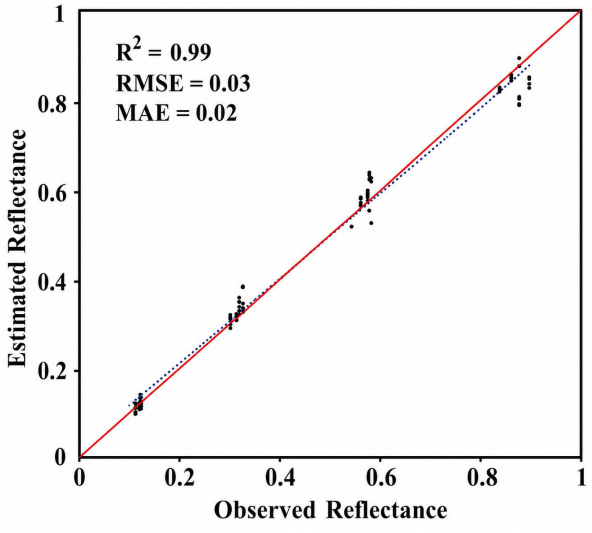


Fig. 11. Performance of the proposed method under cloudy conditions.

small error metrics, confirming the stability of canopy greenness estimation over time. The SR also demonstrated strong agreement between UAV- and SVC-derived values, while GNDVI exhibited slightly lower but still robust correlations, reflecting its higher sensitivity to illumination and spectral variability. Despite this sensitivity, the consistent behavior of GNDVI across crops and stages confirms the effectiveness of the proposed calibration under varying conditions.

The observed stability of vegetation indices across different crop types and phenological stages highlights the ability of the proposed method to maintain reflectance integrity over time. This is particularly important for monitoring crop vigor, canopy structure, and temporal dynamics using UAV-based multispectral imagery. The close alignment between SVC-derived and M3M-derived vegetation indices demonstrates that the calibrated UAV reflectance data can be confidently used for operational crop monitoring and time-series analysis.

C. Performance Evaluation Under Cloudy Conditions

For evaluation under cloudy conditions, the proposed method was tested using UAV images acquired under different cloudy conditions during 2025 and 2026. The variation in cloud thickness across these datasets provides a comprehensive assessment of model performance under variable illumination conditions. The cloudy-condition experiments were conducted under fully overcast sky conditions dominated by cumulus and stratocumulus clouds. Although the sky remained cloud-covered during the acquisition period, variations in cloud thickness and movement produced changes in diffuse illumination, providing a suitable condition for evaluating the temporal stability of the proposed calibration method.

As shown in Fig. 11, the proposed method achieved very high accuracy, with R^2 reaching approximately 0.99, RMSE around 0.029, and MAE approximately 0.020. This strong performance is likely related to the reduced risk of panel saturation, particularly in the green band, under cloudy conditions.

In contrast, the traditional ELM showed lower performance under the same conditions, with $R^2 = 0.90$ and RMSE = 0.1. The higher accuracy observed under fully overcast conditions was partly related to the reduced saturation of the 82% reference panel under diffuse illumination. Therefore, the observed performance reflects both the reduced panel saturation and the capability of the proposed method to effectively handle panel saturation while maintaining robust calibration performance.

D. Impact of Panel Saturation and Robust Calibration Strategy

In the proposed calibration workflow, four reflectance panels were initially used to estimate the camera gain and offset for each spectral band. Under certain illumination conditions, partial saturation was observed in the brightest panel (82% reflectance panel), with DN values approaching the upper digital range of the sensor (approximately 65 000) in the Green and Red bands. When the saturated panel was included in the calibration process, the estimated coefficients exhibited unrealistic negative intercepts, particularly in the Green band. This resulted in systematic underestimation of Green and Red reflectance values associated with the brightest panel. To address this issue, a saturation check was applied to the calibration panels, and any panel with DN values close to the sensor digital limit was excluded from the calibration process. The calibration coefficients were then recomputed using the remaining three unsaturated panels. This revised three-panel calibration produced a more physically consistent relationship between DN and reflectance, effectively removing the bias observed in the Green and Red bands. As a result, the agreement between camera-derived reflectance and reference measurements improved substantially, achieving an RMSE of approximately 0.04 and an R^2 of 0.97. These findings demonstrate that, in practical UAV operations, excluding saturated panels and recalibrating using fewer but reliable reference targets provides a more robust solution than forcing a regression through biased measurements. This approach significantly improves the radiometric quality and reliability of the final reflectance products.

Although removing one saturated reference panel reduces the redundancy of the calibration system, the calibration model remains solvable with the remaining reference panels. Retaining a saturated panel would bias the parameter estimation because its DN values no longer represent the true radiometric response of the sensor. Consequently, excluding saturated panels leads to more reliable calibration despite the reduced redundancy, and the experimental results confirm that the proposed method maintains stable calibration accuracy under these conditions. For consistency, the conventional ELM was also reevaluated after excluding the saturated reference panel. The calibration accuracy improved slightly ($R^2 = 0.91$, RMSE = 0.10); however, the proposed method still achieved substantially higher accuracy ($R^2 = 0.96$, RMSE = 0.04), indicating that its superior performance is maintained even after removing saturated observations. Finally, for saturated reference panels, the saturated DN values were not used directly in the calibration. Instead, the effective irradiance of the saturated panel was reconstructed using the effective

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TABLE V
SENSITIVITY OF THE MODEL TO SPECTRAL LIBRARY UNCERTAINTY

± Change in Reflectance (%)	R^2	RMSE	MAE
±10%	0.93	0.07	0.07
±20%	0.76	2.90	2.70
±30%	0.48	4.11	4.01

irradiance estimated from the remaining unsaturated reference panels of the same spectral band, weighted by their certified spectral-library reflectance values

$$\hat{I}^c_{\lambda,\text{sat}} = \frac{\sum_{i\in\text{unsat}} I^c_{\lambda,i} R^{\text{library}}_{\lambda,i}}{\sum_{i\in\text{unsat}} R^{\text{library}}_{\lambda,i}}$$

where $I^c_{\lambda,i}$ is the effective irradiance estimated for the ith unsaturated reference panel based on (13) and (14), and $R^{\text{library}}_{\lambda,i}$ is its corresponding certified spectral-library reflectance. This strategy avoids using clipped DN values while preserving the spectral-library information of the saturated reference panel. In the panel evaluation, it achieved an R^2 of approximately 0.98 with an RMSE of approximately 0.03. In this study, all calibration panels were carefully cleaned before each experiment, and no visible contamination or soiling was observed during image acquisition. However, any contamination, weathering, or surface degradation may alter the certified panel reflectance and introduce discrepancies with the manufacturer's spectral library. As demonstrated by the sensitivity analysis, such deviations in the reference reflectance can directly affect the calibration accuracy.

E. Sensitivity of the Model to Spectral Library Uncertainty

Since the proposed model relies on spectral library reflectance values as reference inputs, it is important to evaluate its sensitivity to potential uncertainties in these values. To this end, a controlled perturbation analysis was conducted.

In this experiment, the reflectance values of the calibration panels were randomly perturbed by ±10%, ±20%, and ±30% to simulate deviations from the true reference values. These perturbations were applied independently to each panel to represent realistic uncertainty conditions.

For each perturbation level, the experiment was repeated 100 times using random variations, and the model performance was evaluated using R^2, RMSE, and MAE.

As shown in Table V, the model is relatively robust to small deviations, with only a slight decrease in performance at ±10%. However, as the magnitude of perturbation increases, the model performance degrades significantly. At ±20%, both RMSE and MAE increase substantially, while R^2 decreases notably. At ±30%, the degradation becomes severe, with high error values and a significant drop in R^2.

This behavior is expected, as errors in the reference reflectance values directly propagate into the estimation of calibration coefficients and the reconstructed irradiance. These findings indicate that while the proposed method is stable under small uncertainties, its accuracy is highly sensitive to larger deviations. Therefore, the accuracy of the spectral library data is critically important, and the use of reliable and well-calibrated spectral reference data is essential for achieving accurate results.

F. Comparison With Conventional Calibration Approaches and Future Work

Conventional calibration approaches, such as the traditional ELM, rely on simple linear relationships between image DNs and surface reflectance and generally assume constant illumination conditions during image acquisition. As demonstrated by the results of this study, this assumption limits calibration performance under realistic field conditions where illumination varies over time. Using the traditional ELM, the reflectance estimates derived from the M3M exhibited a weaker agreement with SVC spectrometer measurements, with an overall coefficient of determination of $R^2 = 0.88$ (see Fig. 10). The associated error metrics (RMSE = 0.12 and MAE = 0.11) further indicate larger residual errors compared with the proposed physics-based calibration method. In addition, the fit regression line obtained using the traditional ELM deviates below the 1:1 line, revealing a slight systematic underestimation of surface reflectance. In contrast, the proposed physics-based calibration approach achieved a substantially stronger agreement with SVC measurements, with an R^2 value of 0.96 and significantly lower errors (RMSE = 0.04 and MAE = 0.05), as shown in Fig. 8. The fit regression closely overlaps the 1:1 line, indicating reduced bias and improved radiometric accuracy. The near-perfect linearity observed in the RE and NIR bands ($R^2 = 1.00$), together with the strong performance of the visible bands, highlights the benefit of incorporating sensor-specific spectral response information and estimating effective irradiance within the calibration framework.

The improvement in radiometric calibration is also reflected in vegetation index estimation. Compared with the conventional ELM, the proposed method consistently produced vegetation indices with higher agreement with the SVC measurements. For NDVI, the coefficient of determination increased from $R^2 = 0.89$ to $R^2 = 0.94$, while the RMSE decreased from 0.07 to 0.05. Similarly, for SR, the R^2 value increased from 0.87 to 0.93, with the RMSE decreasing from 0.08 to 0.05. GNDVI also showed an improvement in correlation, with the R^2 value increasing from 0.78 to 0.83, while maintaining a similar RMSE of 0.12. These improvements were statistically significant according to paired t-tests ($p < 0.05$), confirming that the proposed calibration method provides a meaningful improvement over the conventional ELM.

To distinguish the effect of the weighting strategy from the proposed effective irradiance formulation, an additional comparison was performed using conventional ELM, ELM with WLS, and the complete proposed method. Adding WLS to the conventional ELM produced only a slight improvement ($R^2 = 0.90$, RMSE = 0.10) compared with the conventional ELM ($R^2 = 0.88$, RMSE = 0.12), and this improvement was not statistically significant according to the paired t-test ($p > 0.05$). In contrast, the complete proposed method achieved substantially higher accuracy ($R^2 = 0.96$, RMSE = 0.04), with a statistically significant improvement over the

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conventional ELM ($p < 0.05$). These results indicate that the improvement is mainly associated with the proposed effective irradiance formulation, while WLS serves as a robust optimization strategy for estimating the calibration coefficients. The improvement is not due to the WLS algorithm itself. Rather, the proposed effective irradiance formulation produces a more physically representative radiometric model by explicitly accounting for illumination variability. Consequently, the residuals are dominated by measurement uncertainty rather than model inadequacy, allowing the WLS estimator to determine the calibration coefficients more accurately and robustly. Although (16) has the same mathematical form as the conventional ELM, the proposed method differs by estimating the effective irradiance as a physical parameter and incorporating a WLSs optimization. These additions improve the robustness of the calibration under practical imaging conditions. The ablation study demonstrates that WLS alone provides only a modest improvement over the conventional ELM, whereas the combination of the physical irradiance formulation and WLS yields the highest calibration accuracy. In addition, the sensitivity analysis shows that accurate reference reflectance is critical for reliable calibration, highlighting the importance of maintaining clean and well-characterized calibration panels.

Adjacency scattering was not explicitly modeled in this study. The reflectance formulation assumes that the measured radiance is primarily dominated by the target surface. This assumption is generally valid for UAV imagery due to the low flight altitude and high spatial resolution, which reduce contributions from surrounding areas. However, in heterogeneous scenes, adjacency effects may introduce additional radiance from neighboring surfaces and affect reflectance estimates. Compared with the traditional method, the proposed approach partially mitigates these effects through the use of calibration panels and the empirical–physical formulation. Future work will focus on incorporating adjacency correction to further improve the physical consistency of the model.

In practical UAV operations, compared with existing radiometric calibration approaches [3], [4], the proposed method provides a low-cost and efficient solution for reflectance calibration by eliminating the need for additional irradiance sensors. This reduces system complexity, simplifies field deployment, and lowers operational costs. The method is suitable for fast surveys and large-area monitoring, where lightweight systems and rapid data collection are required. It also performs well under variable illumination conditions, such as partly cloudy weather. Hence, the proposed method offers a good balance between accuracy and efficiency, which makes it well suited for low-cost, high-efficiency UAV applications.

The reconstructed effective irradiance is reliable under typical UAV acquisition conditions and moderate illumination variability. However, under rapid illumination changes during longer flights, the accuracy may gradually decrease because the irradiance estimation is directly derived from the reference panel measurements. Therefore, periodic recalibration using reference panels is recommended, particularly for flights longer than approximately 15 min. This approach helps reduce illumination-related errors and maintain stable reflectance accuracy.

Although the proposed method demonstrated robust radiometric performance, a quantitative comparison between the calibration error and the intrinsic sensor image noise was beyond the scope of this study. Such an analysis requires quantitative radiometric noise or signal-to-noise ratio (SNR) specifications, which are not publicly available for the DJI M3M sensor. Future work will focus on independently characterizing the sensor noise and investigating its influence on calibration accuracy and downstream applications. Recent studies have also emphasized the importance of improving radiometric calibration under variable illumination conditions. Cao et al. [7] reported that daily illumination variability remains a major source of uncertainty in UAV reflectance estimation. Daniels et al. [8] demonstrated that multipanel ELM generally provides higher calibration accuracy than single-panel approaches, particularly under overcast conditions, whereas incorporating downwelling light sensor (DLS) measurements provided only limited improvement. Swaminathan et al. [4] improved radiometric consistency by integrating DLS measurements into the calibration workflow, while Francis et al. [27] proposed an automated in-flight calibration framework based on reference target detection. In contrast, the proposed method reconstructs the effective irradiance directly from certified spectral library reflectance values and multiple reference panels, eliminating the need for an additional irradiance sensor while preserving the physical irradiance–reflectance relationship.

Overall, these results demonstrate that the proposed physics-based approach outperforms conventional ELM calibration by reducing systematic underestimation and residual errors. This leads to more consistent and temporally stable reflectance estimates across different surface types and varying illumination conditions, making the method more suitable for multiimage UAV-based remote sensing applications.

V. Conclusion

The proposed method provides a practical and physically consistent approach for UAV reflectance calibration. The results clearly show that the proposed method provides accurate, stable, and reliable reflectance estimates. Strong agreement with SVC field measurements across all spectral bands, crop growth stages, and acquisition times confirms the robustness of the method under varying illumination conditions through effective irradiance estimation.

The proposed approach effectively preserves the spectral relationships required for accurate vegetation index computation, which is essential for UAV-based remote sensing applications. The results further demonstrate that combining the physical irradiance formulation with WLSs provides more robust radiometric calibration than the conventional ELM. Consequently, the traditional ELM produced lower accuracy and higher error values than the proposed framework. The substantial reduction in RMSE and MAE achieved by the proposed method demonstrates its clear improvement over the standard calibration procedure.

These findings confirm that the proposed calibration method is well suited for UAV-based remote sensing and can effectively support quantitative remote sensing using multispectral

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UAV data. Furthermore, the method exhibited stable performance across different crop types and surface conditions and significantly reduced the influence of changing solar angle and illumination intensity during the flight period. This makes the proposed approach highly suitable for diverse UAV-based remote sensing monitoring programs.

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REFERENCES

[1] N. Yin et al., "A review: UAV-based remote sensing," *IOP Conf., Mater. Sci. Eng.*, vol. 490, Apr. 2019, Art. no. 062014.

[2] Y. Zhong et al., "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[3] H. Zhu et al., "Assessing radiometric calibration methods for multispectral UAV imagery and the influence of illumination, flight altitude and flight time on reflectance, vegetation index and inversion of winter wheat AGB and LAI," *Comput. Electron. Agricult.*, vol. 219, Apr. 2024, Art. no. 108821.

[4] V. Swaminathan, J. Alex Thomasson, R. G. Hardin, N. Rajan, and R. Raman, "Radiometric calibration of UAV multispectral images under changing illumination conditions with a downwelling light sensor," *Plant Phenome J.*, vol. 7, no. 1, Dec. 2024, Art. no. e70005, doi: 10.1002/ppj2.70005.

[5] L. Deng, X. Hao, Z. Mao, Y. Yan, J. Sun, and A. Zhang, "A subband radiometric calibration method for UAV-based multispectral remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 8, pp. 2869–2880, Aug. 2018.

[6] Y. Guo, J. Senthilnath, W. Wu, X. Zhang, Z. Zeng, and H. Huang, "Radiometric calibration for multispectral camera of different imaging conditions mounted on a UAV platform," *Sustainability*, vol. 11, no. 4, p. 978, Feb. 2019.

[7] S. Cao, B. Danielson, S. Clare, S. Koenig, C. Campos-Vargas, and A. Sanchez-Azofeifa, "Radiometric calibration assessments for UAS-borne multispectral cameras: Laboratory and field protocols," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 132–145, Mar. 2019, doi: 10.1016/j.isprsjprs.2019.01.016.

[8] L. Daniels, E. Eeckhout, J. Wieme, Y. Dejaegher, K. Audenaert, and W. H. Maes, "Identifying the optimal radiometric calibration method for UAV-based multispectral imaging," *Remote Sens.*, vol. 15, no. 11, p. 2909, Jun. 2023, doi: 10.3390/rs15112909.

[9] E. Honkavaara and E. Khoramshahi, "Radiometric correction of close-range spectral image blocks captured using an unmanned aerial vehicle with a radiometric block adjustment," *Remote Sens.*, vol. 10, no. 2, p. 256, Feb. 2018.

[10] A. Burkart, S. Cogliati, A. Schickling, and U. Rascher, "A novel UAV-based ultra-light weight spectrometer for field spectroscopy," *IEEE Sensors J.*, vol. 14, no. 1, pp. 62–67, Jan. 2014.

[11] T. Hakala et al., "Direct reflectance measurements from drones: Sensor absolute radiometric calibration and system tests for forest reflectance characterization," *Sensors*, vol. 18, no. 5, p. 1417, May 2018.

[12] J. Kelcey and A. Lucieer, "Sensor correction and radiometric calibration of a 6-band multispectral imaging sensor for uav remote sensing," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 1, pp. 393–398, Jul. 2012.

[13] E. Chakhvashvili, B. Siegmann, J. Bendig, and U. Rascher, "Comparison of reflectance calibration workflows for a UAV-mounted multi-camera array system," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 8225–8228.

[14] E. Honkavaara et al., "Processing and assessment of spectrometric, stereoscopic imagery collected using a lightweight UAV spectral camera for precision agriculture," *Remote Sens.*, vol. 5, no. 10, pp. 5006–5039, Oct. 2013.

[15] C. Wang and S. W. Myint, "A simplified empirical line method of radiometric calibration for small unmanned aircraft systems-based remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 1876–1885, May 2015.

[16] F. Gascon, J.-P. Gastellu-Etchegorry, and M.-J. Lefevre, "Radiative transfer model for simulating high-resolution satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 9, pp. 1922–1926, Sep. 2001.

[17] E. Vermote, D. Tanré, J. L. Deuzé, and M. Herman, "Second simulation of a satellite signal in the solar spectrum–vector (6SV)," Dept. Geography, Univ. Maryland, College Park, MD, USA, Tech. Rep., Nov. 2006.

[18] N. Hashimoto, Y. Saito, M. Maki, and K. Homma, "Simulation of reflectance and vegetation indices for unmanned aerial vehicle (UAV) monitoring of paddy fields," *Remote Sens.*, vol. 11, no. 18, p. 2119, Sep. 2019.

[19] J. Xie, Y. Shen, and H. Cen, "Real-time reflectance generation for UAV multispectral imagery using an onboard downwelling spectrometer in varied weather conditions," 2024, *arXiv:2412.19527*.

[20] G. M. Smith and E. J. Milton, "The use of the empirical line method to calibrate remotely sensed data to reflectance," *Int. J. Remote Sens.*, vol. 20, no. 13, pp. 2653–2662, Jan. 1999.

[21] M. S. Moran et al., "A refined empirical line approach for reflectance factor retrieval from Landsat-5 TM and Landsat-7 ETM+," *Remote Sens. Environ.*, vol. 78, nos. 1–2, pp. 71–82, Oct. 2001.

[22] S. Del Pozo, P. Rodríguez-Gonzálvez, D. Hernández-López, and B. Felipe-García, "Vicarious radiometric calibration of a multispectral camera on board an unmanned aerial system," *Remote Sens.*, vol. 6, no. 3, pp. 1918–1937, Feb. 2014.

[23] E. Ben-Dor, F. A. Kruse, A. B. Lefkoff, and A. Banin, "Comparison of three calibration techniques for utilization of GER 63-channel aircraft scanner data," *Photogramm. Eng. Remote Sens.*, vol. 60, pp. 1339–1354, Nov. 1994.

[24] G. W. Staben, K. Pfitzner, R. Bartolo, and A. Lucieer, "Empirical line calibration of WorldView-2 satellite imagery to reflectance data: Using quadratic prediction equations," *Remote Sens. Lett.*, vol. 3, no. 6, pp. 521–530, Nov. 2012.

[25] R. F. Kokaly et al., "USGS spectral library version 7," *U.S. Geological Surv. Data Ser. 1035*, Reston, VA, USA, pp. 1–61, 2017, doi: 10.3133/ds1035.

[26] A. M. Baldridge, S. J. Hook, C. I. Grove, and G. Rivera, "The ASTER spectral library version 2.0," *Remote Sens. Environ.*, vol. 113, no. 4, pp. 711–715, Apr. 2009, doi: 10.1016/j.rse.2008.11.007.

[27] L. J. Francis, L. Gabriel B. Geissler, N. Okole, B. Gipp, C. Stachniss, and R. Heim, "ReflectDetect: A software tool for AprilTag-guided in-flight radiometric calibration for UAV-mounted 2D snapshot multi-camera imagery," *SoftwareX*, vol. 30, May 2025, Art. no. 102150, doi: 10.1016/j.softx.2025.102150.

[28] *DJI Mavic 3 Multispectral User Manual*, Version 1.0, SZ DJI Technology Co., Shenzhen, China, Dec. 2022.

[29] G. Chander, B. L. Markham, and D. L. Helder, "Summary of current radiometric calibration coefficients for Landsat MSS, TM, ETM+, and EO-1 ALI sensors," *Remote Sens. Environ.*, vol. 113, no. 5, pp. 893–903, May 2009, doi: 10.1016/j.rse.2009.01.007.

[30] U.S. Geological Survey, Reston, VA, USA. *Using the USGS Landsat Level-1 Data Product*. Accessed: Oct. 1, 2026. [Online]. Available: https://www.usgs.gov/landsat-missions/using-usgs-landsat-level-1-data-product

[31] R. L. Pearson and L. D. Miller, "Remote mapping of standing crop biomass for estimation of the productivity of the shortgrass Prairie," in *Proc. 8th Int. Symp. Remote Sens. Environ.*, Ann Arbor, MI, USA, 1972, pp. 1357–1381.

[32] J. W. Rouse Jr., R. H. Haas, J. A. Schell, and D. W. Deering, "Monitoring vegetation systems in the great plains with ERTS," in *Proc. 3rd ERTS Symp.*, 1974, pp. 309–317.

[33] A. A. Gitelson, A. Viña, T. J. Arkebauer, D. C. Rundquist, G. Keydan, and B. Leavitt, "Remote estimation of leaf area index and green leaf biomass in maize canopies," *Geophys. Res. Lett.*, vol. 32, no. 8, Mar. 2005, Art. no. L08403.

[34] J. L. Hatfield and J. H. Prueger, "Value of using different vegetative indices to quantify agricultural crop characteristics at different growth stages under varying management practices," *Remote Sens.*, vol. 2, no. 2, pp. 562–578, Feb. 2010.

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[35] C. D. Elvidge and Z. Chen, "Comparison of broad-band and narrow-band red and near-infrared vegetation indices," *Remote Sens. Environ.*, vol. 54, no. 1, pp. 38–48, Oct. 1995.



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